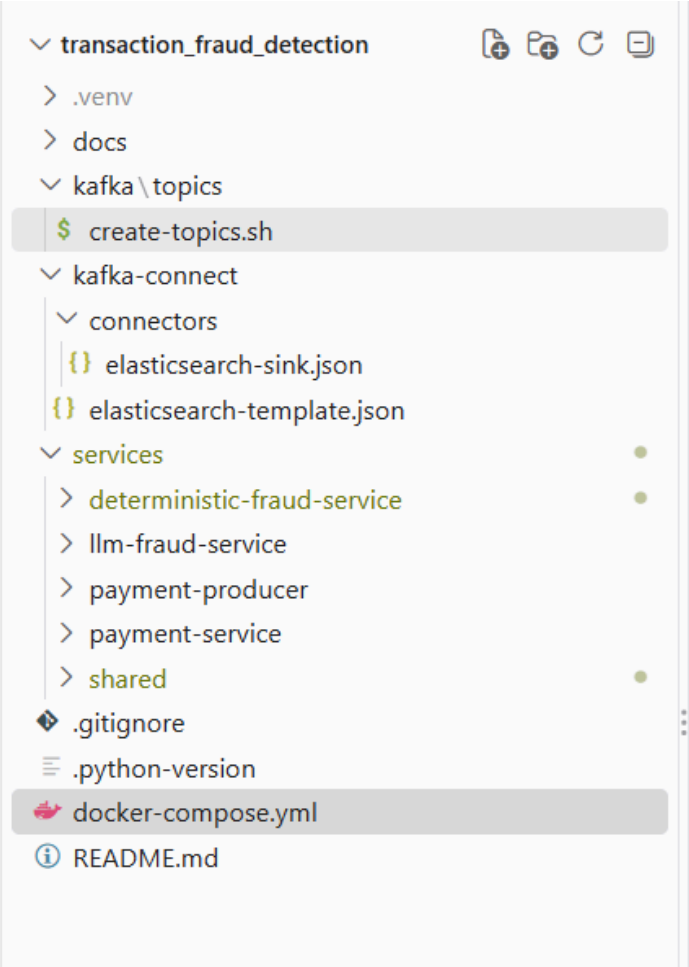
Understanding the source code

This section outlines the project's primary code modules, overall structure, and key configurations, followed by an explanation of how these components are built.

Overall Folder Structure:



docker-compose.yml - It contains the specific configurations to run the below services in containerised environment:

1) Kafka
2) Kafka-connect
3) Redis
4) Elasticsearch
5) Kibana

Additionally the different python fraud services can also be added in this configuration

kafka/topics/create-topics.sh - This script is used to create different kafka topics discussed in design during the startup phase

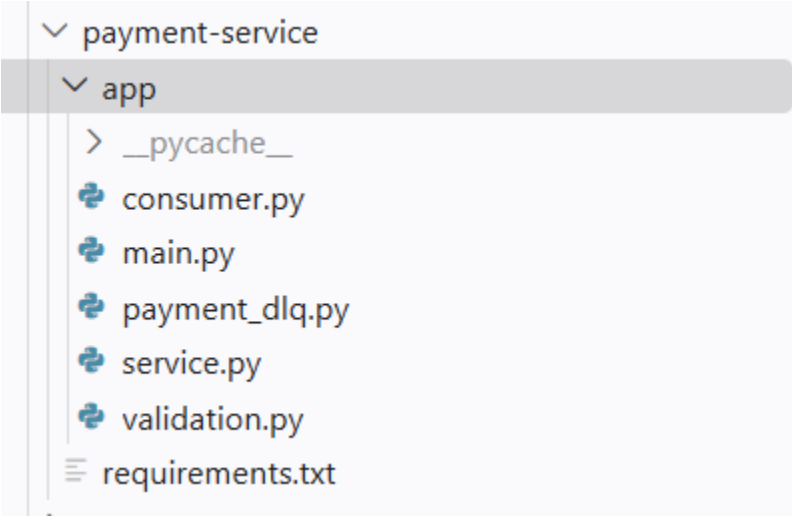
kafka-connect/connectors/elasticsearch-sink.json - This configuration sets up Elasticsearch Sink Connector that streams JSON data from the payment-audit Kafka topic to Elasticsearch.

kafka-connect/elasticsearch-template.json - Elasticsearch Index Template that automatically applies predefined mappings, settings, and field data types to newly created indices that match specific name patterns.

Services folder - Contain the python implementation of all the different fraud detection services discussed in the design and I will briefly touch base on the main modules of the services in the next section.

Code structure of payment-service:

This is the payment ingestion service which does the payment ingestion from Kafka, performs validation and publishes payment results back to Kafka for downstream fraud detection services.



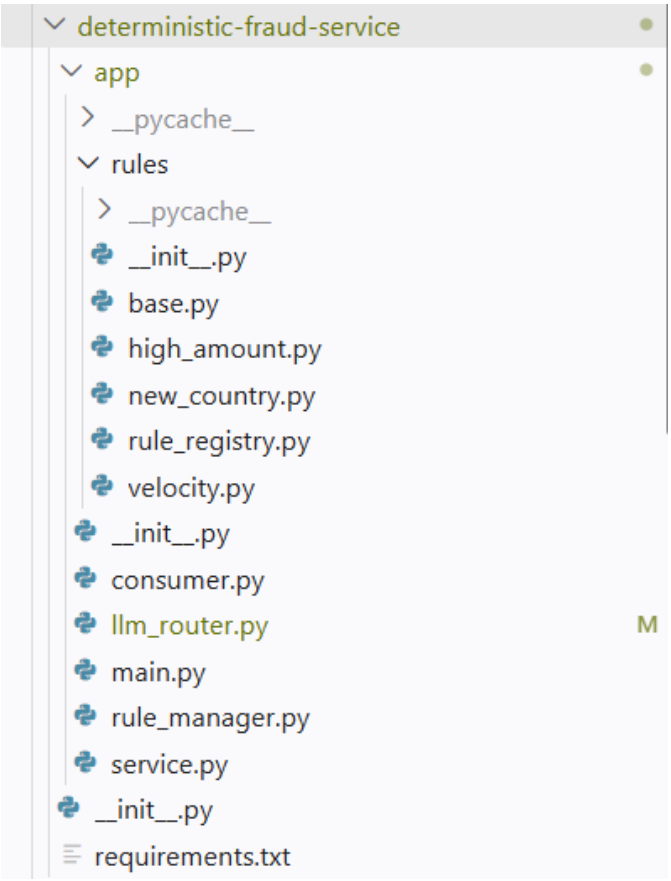
consumer.py - It implements the logic to consume payment events from kafka using consumer group and maintains the logic to send failed messages to dlq.

service.py - Implements the business logic for running the payment service process

validation.py - Payment event validation file.

Code structure of deterministic-fraud-service:

This is the rule based payment fraud detection service for generating fraud alerts. For any ambiguous fraud analysis outcome it flags payment events for further llm fraud service review.



consumer.py - Implements the logic to consume payment results from Kafka topic and send it for fraud detection process

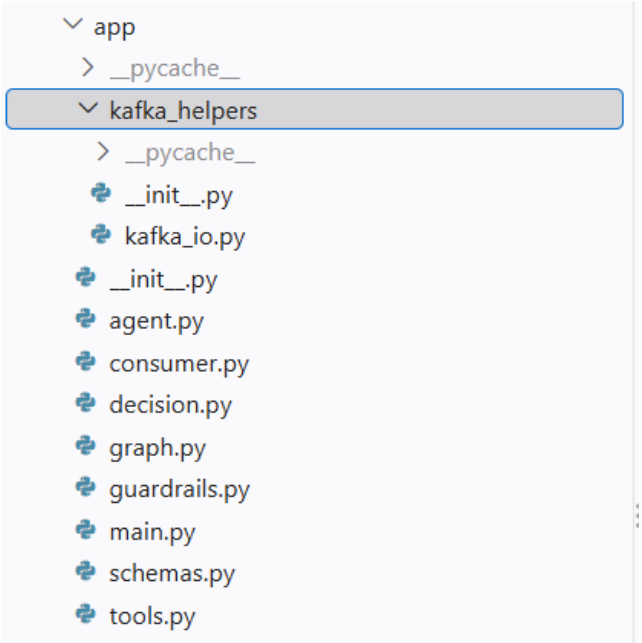
service.py - Implements the business logic for running the rule based fraud detection process

rules: Contains implementation code for different fraud rules like velocity, new country and high payment amount. Rule manager registers and executes these rules

llm_router.py - Implements the policy decision for the analysed payment event to be flagged for llm review.

Code structure of llm-fraud-service:

This service using LLM based fraud detection analyzes those ambiguous payment events for which rule based fraud service fails to produce clear outcomes.



consumer.py - It implements the logic to consume flagged payment events for llm review from kafka using consumer group and send it for llm based fraud detection process

service.py - Implements the business logic for running the llm based fraud detection process.

graph.py - This file contains business logic implementation of different langgraph nodes and edges.

guardrails.py - Implements the different input and output validation for the llm call and is used inside the langgraph flow.

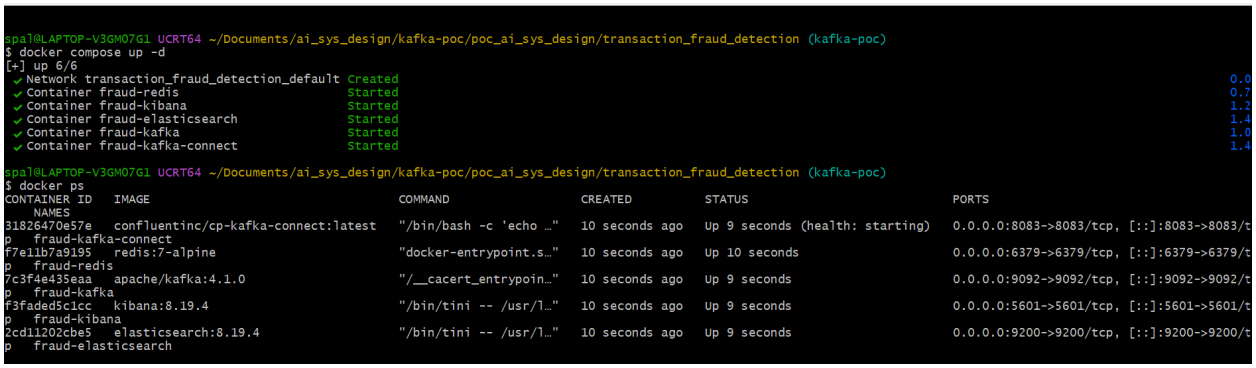
tools.py - Agent tools to retrieve customer context like transaction history and user profile from Redis store

agent.py - Configuration of langchain deepagent harness.

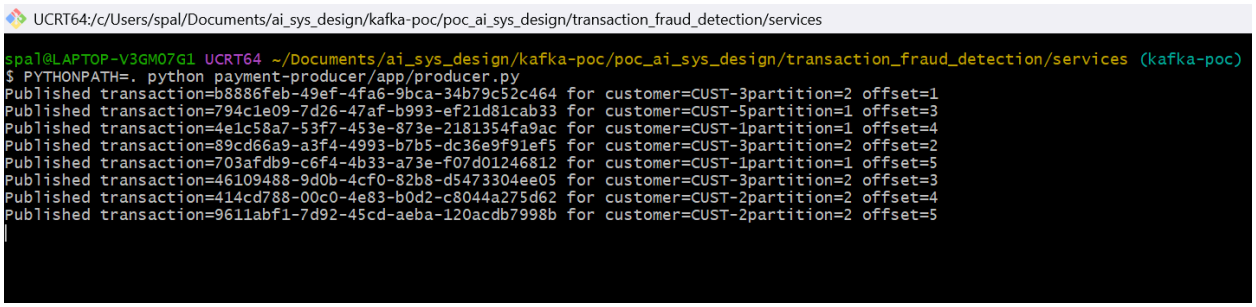
Running the project and results

The steps to run the project are mentioned in the **readme** file, sharing below the execution results.

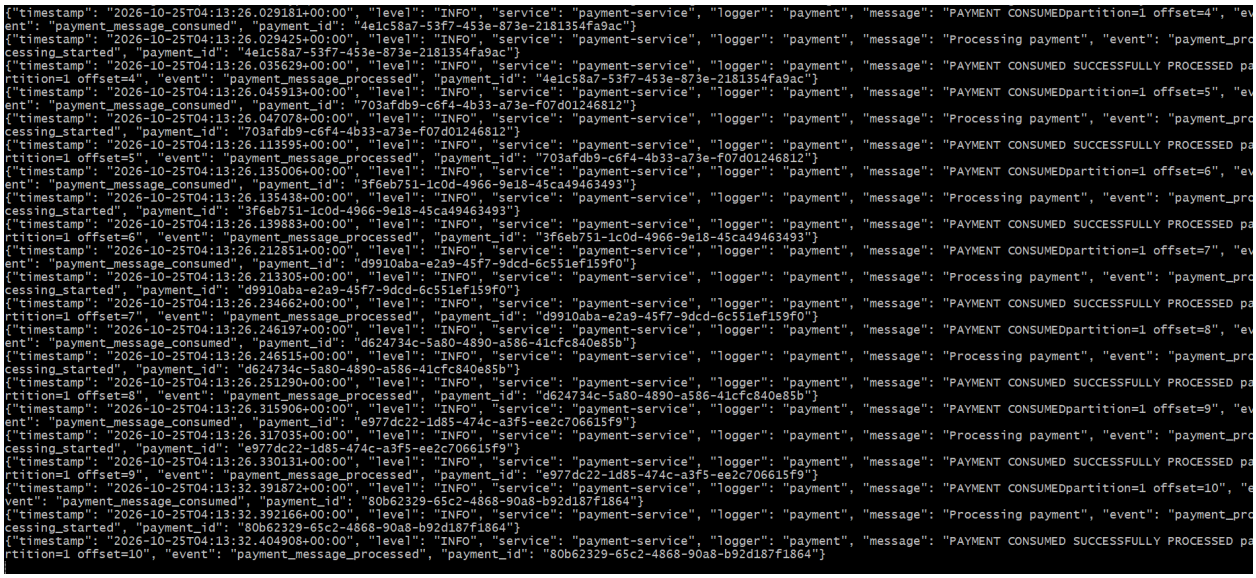
Running Containers:



Payment simulator process logs:

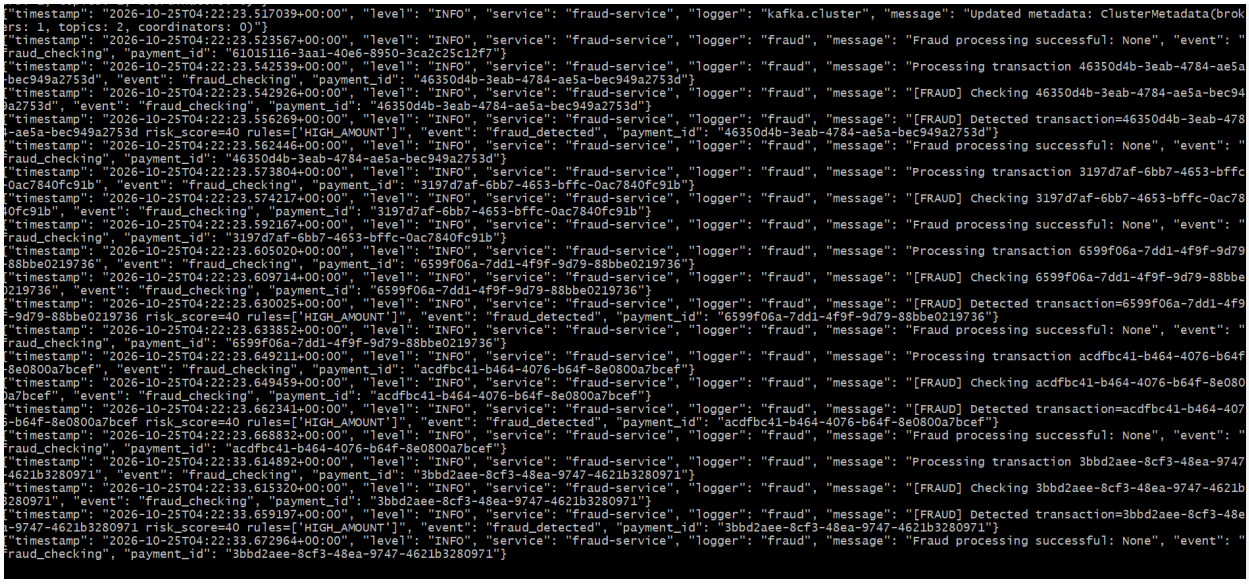


Payment Ingestion service logs:



```
{"timestamp": "2026-10-25T04:13:26.029181+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMEDpartition=1 offset=4", "ev
ent": "payment_message_consumed", "payment_id": "4e1c58a7-53f7-453e-873e-2181354fa9ac"}
{"timestamp": "2026-10-25T04:13:26.029425+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "Processing payment", "event": "payment_pro
cessing_started", "payment_id": "4e1c58a7-53f7-453e-873e-2181354fa9ac"}
{"timestamp": "2026-10-25T04:13:26.035629+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMED SUCCESSFULLY PROCESSED pa
rtition=1 offset=4", "event": "payment_message_processed", "payment_id": "4e1c58a7-53f7-453e-873e-2181354fa9ac"}
{"timestamp": "2026-10-25T04:13:26.045913+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMEDpartition=1 offset=5", "ev
ent": "payment_message_consumed", "payment_id": "703afdb9-c6f4-4b33-a73e-f07d01246812"}
{"timestamp": "2026-10-25T04:13:26.047078+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "Processing payment", "event": "payment_pro
cessing_started", "payment_id": "703afdb9-c6f4-4b33-a73e-f07d01246812"}
{"timestamp": "2026-10-25T04:13:26.113595+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMED SUCCESSFULLY PROCESSED pa
rtition=1 offset=5", "event": "payment_message_processed", "payment_id": "703afdb9-c6f4-4b33-a73e-f07d01246812"}
{"timestamp": "2026-10-25T04:13:26.135006+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMEDpartition=1 offset=6", "ev
ent": "payment_message_consumed", "payment_id": "3f6eb751-1c0d-4966-9e18-45ca49463493"}
{"timestamp": "2026-10-25T04:13:26.135438+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "Processing payment", "event": "payment_pro
cessing_started", "payment_id": "3f6eb751-1c0d-4966-9e18-45ca49463493"}
{"timestamp": "2026-10-25T04:13:26.139883+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMED SUCCESSFULLY PROCESSED pa
rtition=1 offset=6", "event": "payment_message_processed", "payment_id": "3f6eb751-1c0d-4966-9e18-45ca49463493"}
{"timestamp": "2026-10-25T04:13:26.212851+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMEDpartition=1 offset=7", "ev
ent": "payment_message_consumed", "payment_id": "d9910aba-e2a9-45f7-9dcd-6c551ef159f0"}
{"timestamp": "2026-10-25T04:13:26.213305+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "Processing payment", "event": "payment_pro
cessing_started", "payment_id": "d9910aba-e2a9-45f7-9dcd-6c551ef159f0"}
{"timestamp": "2026-10-25T04:13:26.234662+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMED SUCCESSFULLY PROCESSED pa
rtition=1 offset=7", "event": "payment_message_processed", "payment_id": "d9910aba-e2a9-45f7-9dcd-6c551ef159f0"}
{"timestamp": "2026-10-25T04:13:26.246197+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMEDpartition=1 offset=8", "ev
ent": "payment_message_consumed", "payment_id": "d624734c-5a80-4890-a586-41cfc840e85b"}
{"timestamp": "2026-10-25T04:13:26.246515+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "Processing payment", "event": "payment_pro
cessing_started", "payment_id": "d624734c-5a80-4890-a586-41cfc840e85b"}
{"timestamp": "2026-10-25T04:13:26.251290+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMED SUCCESSFULLY PROCESSED pa
rtition=1 offset=8", "event": "payment_message_processed", "payment_id": "d624734c-5a80-4890-a586-41cfc840e85b"}
{"timestamp": "2026-10-25T04:13:26.315906+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMEDpartition=1 offset=9", "ev
ent": "payment_message_consumed", "payment_id": "e977dc22-1d85-474c-a3f5-ee2c706615f9"}
{"timestamp": "2026-10-25T04:13:26.317035+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "Processing payment", "event": "payment_pro
cessing_started", "payment_id": "e977dc22-1d85-474c-a3f5-ee2c706615f9"}
{"timestamp": "2026-10-25T04:13:26.330131+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMED SUCCESSFULLY PROCESSED pa
rtition=1 offset=9", "event": "payment_message_processed", "payment_id": "e977dc22-1d85-474c-a3f5-ee2c706615f9"}
{"timestamp": "2026-10-25T04:13:32.391872+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMEDpartition=1 offset=10", "e
vent": "payment_message_consumed", "payment_id": "80b62329-65c2-4868-90a8-b92d187f1864"}
{"timestamp": "2026-10-25T04:13:32.392166+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "Processing payment", "event": "payment_pro
cessing_started", "payment_id": "80b62329-65c2-4868-90a8-b92d187f1864"}
{"timestamp": "2026-10-25T04:13:32.404908+00:00", "level": "INFO", "service": "payment-service", "logger": "payment", "message": "PAYMENT CONSUMED SUCCESSFULLY PROCESSED pa
rtition=1 offset=10", "event": "payment_message_processed", "payment_id": "80b62329-65c2-4868-90a8-b92d187f1864"}
```


Rule based fraud detection service logs:

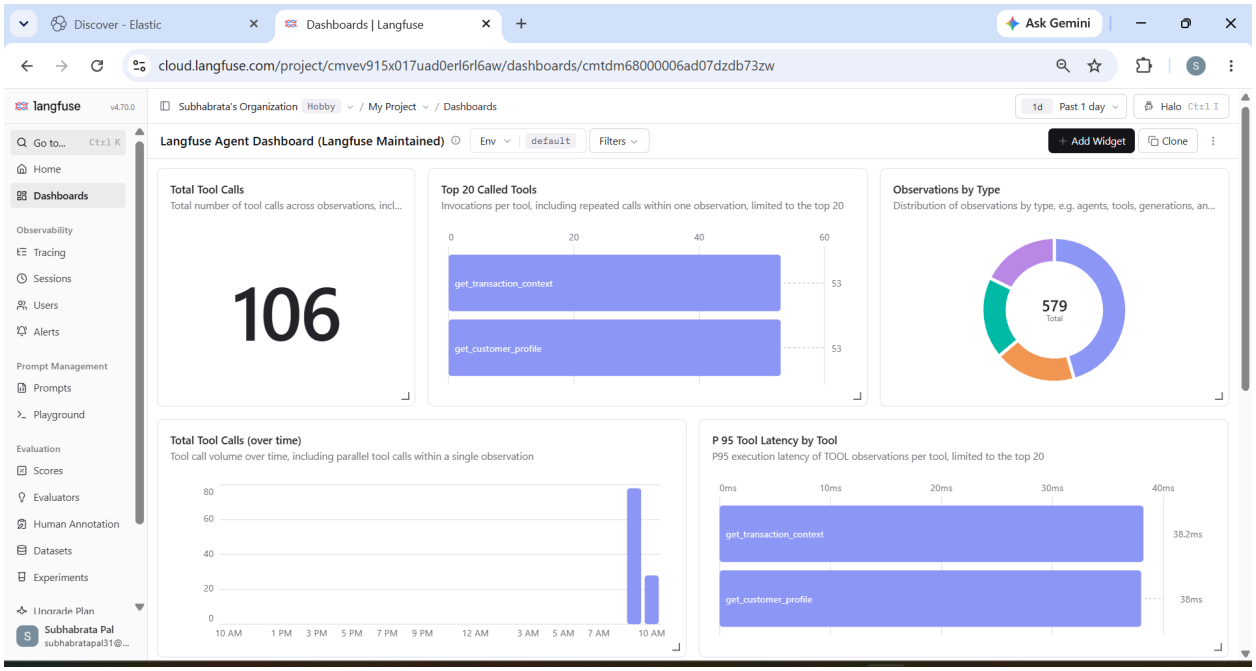


```
{"timestamp": "2026-10-25T04:22:23.517039+00:00", "level": "INFO", "service": "fraud-service", "logger": "kafka.cluster", "message": "Updated metadata: ClusterMetadata(brok
ers: 1, topics: 2, coordinators: 0)"}
{"timestamp": "2026-10-25T04:22:23.523567+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Fraud processing successful: None", "event": "
fraud_checking", "payment_id": "61015116-3aa1-40e6-8950-3ca2c25c12f7"}
{"timestamp": "2026-10-25T04:22:23.542539+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Processing transaction 46350d4b-3eab-4784-ae5a
-bec949a2753d", "event": "fraud_checking", "payment_id": "46350d4b-3eab-4784-ae5a-bec949a2753d"}
{"timestamp": "2026-10-25T04:22:23.542926+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Checking 46350d4b-3eab-4784-ae5a-bec94
9a2753d", "event": "fraud_checking", "payment_id": "46350d4b-3eab-4784-ae5a-bec949a2753d"}
{"timestamp": "2026-10-25T04:22:23.556269+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Detected transaction=46350d4b-3eab-478
4-ae5a-bec949a2753d risk_score=40 rules=['HIGH_AMOUNT']", "event": "fraud_detected", "payment_id": "46350d4b-3eab-4784-ae5a-bec949a2753d"}
{"timestamp": "2026-10-25T04:22:23.562446+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Fraud processing successful: None", "event": "
fraud_checking", "payment_id": "46350d4b-3eab-4784-ae5a-bec949a2753d"}
{"timestamp": "2026-10-25T04:22:23.573804+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Processing transaction 3197d7af-6bb7-4653-bffc
-0ac7840fc91b", "event": "fraud_checking", "payment_id": "3197d7af-6bb7-4653-bffc-0ac7840fc91b"}
{"timestamp": "2026-10-25T04:22:23.574217+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Checking 3197d7af-6bb7-4653-bffc-0ac78
40fc91b", "event": "fraud_checking", "payment_id": "3197d7af-6bb7-4653-bffc-0ac7840fc91b"}
{"timestamp": "2026-10-25T04:22:23.592167+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Fraud processing successful: None", "event": "
fraud_checking", "payment_id": "3197d7af-6bb7-4653-bffc-0ac7840fc91b"}
{"timestamp": "2026-10-25T04:22:23.605020+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Processing transaction 6599f06a-7dd1-4f9f-9d79
-88bbe0219736", "event": "fraud_checking", "payment_id": "6599f06a-7dd1-4f9f-9d79-88bbe0219736"}
{"timestamp": "2026-10-25T04:22:23.609714+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Checking 6599f06a-7dd1-4f9f-9d79-88bbe
0219736", "event": "fraud_checking", "payment_id": "6599f06a-7dd1-4f9f-9d79-88bbe0219736"}
{"timestamp": "2026-10-25T04:22:23.630025+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Detected transaction=6599f06a-7dd1-4f9
f-9d79-88bbe0219736 risk_score=40 rules=['HIGH_AMOUNT']", "event": "fraud_detected", "payment_id": "6599f06a-7dd1-4f9f-9d79-88bbe0219736"}
{"timestamp": "2026-10-25T04:22:23.633852+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Fraud processing successful: None", "event": "
fraud_checking", "payment_id": "6599f06a-7dd1-4f9f-9d79-88bbe0219736"}
{"timestamp": "2026-10-25T04:22:23.649211+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Processing transaction acdfbc41-b464-4076-b64f
-8e0800a7bcef", "event": "fraud_checking", "payment_id": "acdfbc41-b464-4076-b64f-8e0800a7bcef"}
{"timestamp": "2026-10-25T04:22:23.649459+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Checking acdfbc41-b464-4076-b64f-8e080
0a7bcef", "event": "fraud_checking", "payment_id": "acdfbc41-b464-4076-b64f-8e0800a7bcef"}
{"timestamp": "2026-10-25T04:22:23.662341+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Detected transaction=acdfbc41-b464-407
6-b64f-8e0800a7bcef risk_score=40 rules=['HIGH_AMOUNT']", "event": "fraud_detected", "payment_id": "acdfbc41-b464-4076-b64f-8e0800a7bcef"}
{"timestamp": "2026-10-25T04:22:23.668832+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Fraud processing successful: None", "event": "
fraud_checking", "payment_id": "acdfbc41-b464-4076-b64f-8e0800a7bcef"}
{"timestamp": "2026-10-25T04:22:33.614892+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Processing transaction 3bbd2aee-8cf3-48ea-9747
-4621b3280971", "event": "fraud_checking", "payment_id": "3bbd2aee-8cf3-48ea-9747-4621b3280971"}
{"timestamp": "2026-10-25T04:22:33.615320+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Checking 3bbd2aee-8cf3-48ea-9747-4621b
3280971", "event": "fraud_checking", "payment_id": "3bbd2aee-8cf3-48ea-9747-4621b3280971"}
{"timestamp": "2026-10-25T04:22:33.659197+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "[FRAUD] Detected transaction=3bbd2aee-8cf3-48e
a-9747-4621b3280971 risk_score=40 rules=['HIGH_AMOUNT']", "event": "fraud_detected", "payment_id": "3bbd2aee-8cf3-48ea-9747-4621b3280971"}
{"timestamp": "2026-10-25T04:22:33.672964+00:00", "level": "INFO", "service": "fraud-service", "logger": "fraud", "message": "Fraud processing successful: None", "event": "
fraud_checking", "payment_id": "3bbd2aee-8cf3-48ea-9747-4621b3280971"}
```

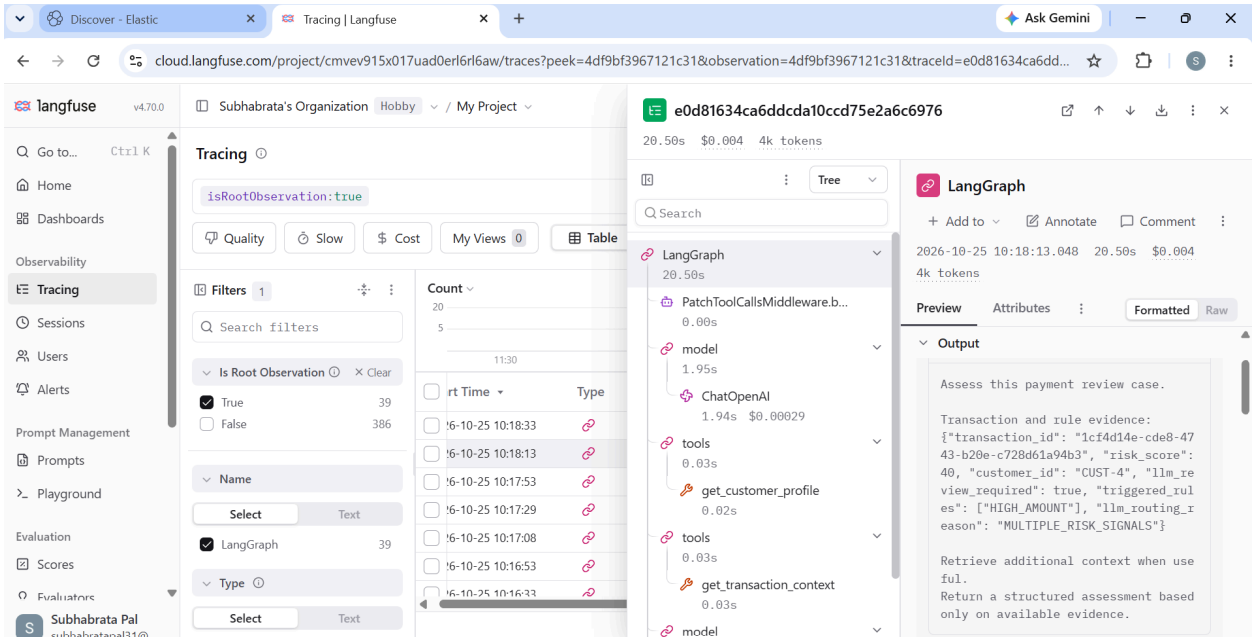
LLM based fraud detection service logs:


```
{"timestamp": "2026-10-25T04:35:24.638512+00:00", "level": "INFO", "service": "llm-fraud-service", "logger": "kafka.cluster", "message": "Updated metadata: ClusterMetadata(
brokers: 1, topics: 1, coordinators: 1)"}
{"timestamp": "2026-10-25T04:35:28.535568+00:00", "level": "INFO", "service": "llm-fraud-service", "logger": "httpx2", "message": "HTTP Request: POST https://api.openai.com
/v1/chat/completions \"HTTP/1.1 200 OK\""}
{"timestamp": "2026-10-25T04:35:56.999971+00:00", "level": "INFO", "service": "llm-fraud-service", "logger": "httpx2", "message": "HTTP Request: POST https://api.openai.com
/v1/chat/completions \"HTTP/1.1 200 OK\""}
{"timestamp": "2026-10-25T04:35:57.012571+00:00", "level": "INFO", "service": "llm-fraud-service", "logger": "kafka.cluster", "message": "Updated metadata: ClusterMetadata(
brokers: 1, topics: 1, coordinators: 0)"}
{"timestamp": "2026-10-25T04:35:57.034736+00:00", "level": "INFO", "service": "llm-fraud-service", "logger": "llm-fraud-service", "message": "Fraud assessment completed tra
nsaction_id=003448ba-644b-4f41-86f0-3e7c3ca075a3 status=FALLBACK_REVIEW"}
{"timestamp": "2026-10-25T04:35:59.115903+00:00", "level": "INFO", "service": "llm-fraud-service", "logger": "httpx2", "message": "HTTP Request: POST https://api.openai.com
/v1/chat/completions \"HTTP/1.1 200 OK\""}
{"timestamp": "2026-10-25T04:36:17.856483+00:00", "level": "INFO", "service": "llm-fraud-service", "logger": "httpx2", "message": "HTTP Request: POST https://api.openai.com
```

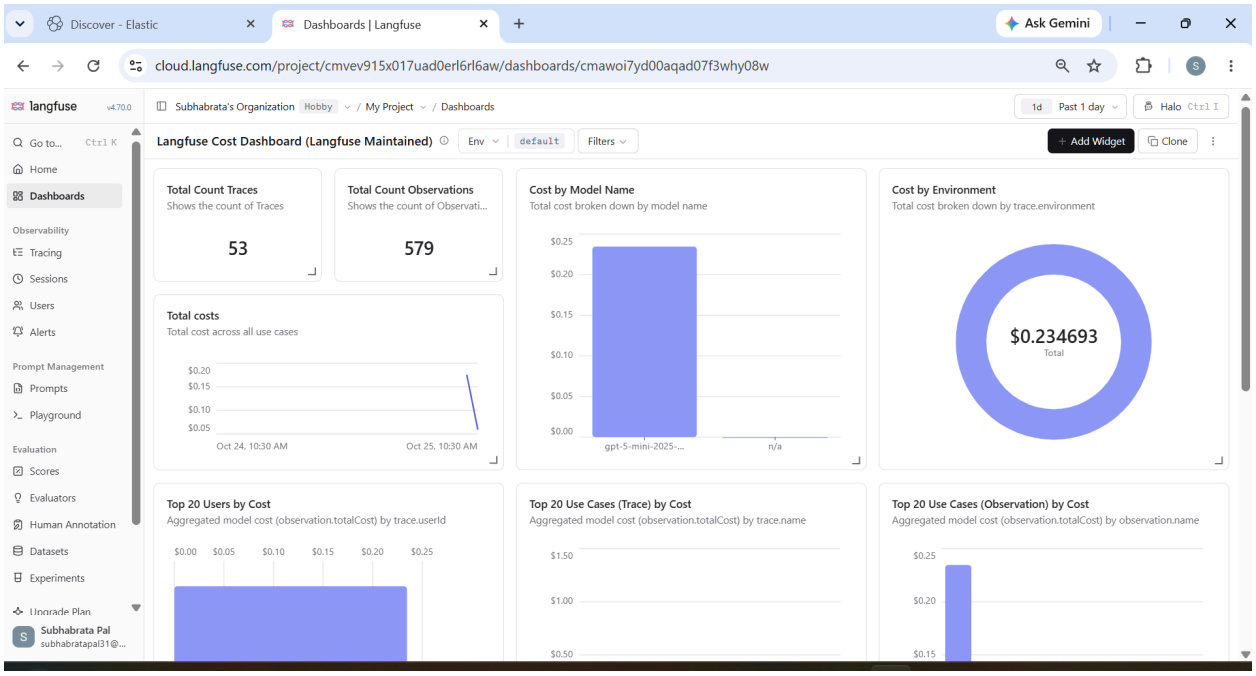

Langfuse Agent Dashboard:



Langfuse Traces:



Token Cost Dashboard:



Elasticsearch Audit Logs:

